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# Interim Market Commentary

***The Implications of the Debt Cycle***

***Leveraged Versus Unleveraged Businesses***

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## *The Implications of the Debt Cycle*

The late economist, Hyman Minsky, had a very interesting model pertaining to speculation with debt. He basically divided the sequence into three speculative phases. In the first phase, there technically isn't any speculation with debt. Entrepreneurs borrow money to undertake projects and the income on the projects is usually sufficient to amortize the principal of the debt and pay the interest. That's very stable. In the second phase, which becomes somewhat more speculative, the earnings in the project are sufficient to service the debt but not to meaningfully amortize the principal. In the last phase, which is the speculative phase, the earnings of the projects undertaken are neither sufficient to amortize principal nor to service the debt; the entire undertaking is based on the hope – and usually it's a fallacy – that the project will appreciate in value, eventually be sold, and then by its appreciation efface the debt. Usually the speculative phase is the beginning of trouble in the debt cycle.

I have often wondered why it is that this cycle repeats itself again and again. One would think that people would learn from history, but they don't seem to. This cycle has bearing upon every participant in the economy and every participant in the equity market, even if one doesn't actually borrow money. For example, let us examine the S&P 500, which has an assets-to-equity ratio of something in excess of 2.6x. There are a not insignificant number of leveraged companies in the S&P 500. In the recent credit crisis, some companies in the S&P were not leveraged at all, some were reasonably leveraged, while others had great difficulty servicing their debt. Examples of companies in this last category were General Electric and Bank of America. It is arguable that without government intervention they might not have been able to refund their debt and all sorts of catastrophes might have occurred, even beyond the catastrophes that did occur.

Therefore, the question arises, if one is simply an equity investor, is it necessary as a passive equity investor to undertake the leverage that is contained in the S&P 500? Another dimension of this problem is that many of the companies in the S&P, because of the availability of credit at times, are undertaking not very profitable projects. These companies might have internal rates of return on projects of 3% to 5%. Properly leveraged, they might actually return a double-digit quantity. These companies really entail risk for the passive investor who does not wish to be leveraged. So I pose another question – is it possible to have an acceptable rate of return and at the same time reduce the leverage and still remain a passive investor?

A third dimension to all of this is that much effort is devoted to managing the inherent variability of the index. This is done through swaps, through collars, or various kinds of option strategies. But perhaps there's a simpler way to manage it. It is certainly arguable that much of the price variability of the S&P itself on a daily basis is ultimately a function of the leverage that the various companies collectively undertake.

I read something recently which has nothing to do with finance, but which gave me a thought in this direction. I read something about the training strategy of the Imperial Japanese Navy in the years before the Second World War. The Japanese Navy pilot training program was far more advanced than the U.S. Navy pilot training program. The candidates selected for pilot training were usually themselves a higher order of candidate. To qualify as a Naval aviator in the Japanese Navy – Nihon Teikoku Kaigun – one needed 700 hours of flying time. A U.S. Navy pilot needed 305 hours at that time. The Imperial Navy pilot training program lasted anywhere from 50-64 months, meaning that it certainly in all cases exceeded four years, and in many cases exceeded five years. The variability was the function of the experience level of the military personnel once they were selected for Naval aviation. In contrast, the U.S. Navy pilot training program was 18 months long. So the Japanese aviators were far more skilled than U.S. aviators, in addition to which, the Japanese Naval aviators, or at least some of them, actually had combat aviation experience as Japanese Air Force aviators, whereas the American pilots, not being exposed to military combat before the war, essentially had no combat experience.

The standards of the Japanese Navy pilot training were so rigorous that the program could only graduate 100 pilots per year. The Japanese program was designed to produce the best aviators, but of course, the higher the standard, the fewer number of pilots will qualify. The Japanese, achieving victory at Pearl Harbor, actually lost 29 pilots on that day. So basically, the pilot losses were going to absorb 29 percent of the incoming graduating class of pilots. It’s fairly obvious that a normal quantity of pilot wastage from combat and other accidents was going to be such that the Japanese Navy undertaking military operations against the United States would not be able to replace its combat aviators. It wasn’t very many years before the U.S., which had much lower standards, was graduating far more aviators. So the Japanese had no alternative but to gradually lower their own standards to produce an equivalent number of aviators. Towards the end of the war, the number of months required to qualify as a certified combat naval aviator in Japan was four months. All one can learn in four months is enough skill to take off and fly the plane into a ship in a Kamikaze attack, which is what happened. So the Japanese had the best program, but the Americans had the most reasonable program. And the reasonable program was the program that was sufficient for the objective.

Now let’s return to this question of debt. In buying the S&P 500 as an index, one is buying the shares of superbly managed companies that are incredibly disciplined and, generally speaking, do not undertake projects unless they have a high probability of generating a double-digit internal rate of return. And in that index are also many other companies that have extreme amounts of leverage on their balance sheets and actually undertake very low return projects that are only profitable because the cost of debt capital is actually lower than the projected internal rate of return on the project itself. Therefore, one wonders, is it not possible to create an index of the best managed, best disciplined, less leveraged companies – let’s say those with sufficient balance sheet strength and the discipline to consistently earn double-digit returns on equity – and then, outside of the index, leverage those companies?

In other words, let us assume there existed a group of companies with businesses that are not variable, and which earned a return-on-equity on average of 12-plus percent, and let's say those companies as a group had balance sheets that were 10% leveraged. And let's say that this group, as an index, was leveraged 30% by an investor. Therefore, looking at a portfolio so structured, the total assets to equity ratio would be 1.4x. One actually would create, even though one is leveraging companies, a less leveraged index. Therefore, the question is, would such an index actually be more stable and would it have a higher rate of return than the actual index itself? And if the answer is yes, what are the consequences for modern portfolio theory if one can actually do something like this?

## *Leveraged Versus Unleveraged Businesses*

Once again I have created an industry that really doesn't exist. Better stated, I've created another way of classifying companies, because the industries in which companies operate are merely one method of categorization. Clearly, one homogeneous characteristic of companies is that they all sell the same product. However, the fact that they sell a given product is not their only characteristic. Another characteristic that they may or may not have in common is the amount of balance sheet debt they're willing to accumulate, or the amount of financial leverage they're willing to employ. Therefore, I have here two portfolios which I've labeled Portfolio I and Portfolio II.

| Portfolio I | Assets-to-Equity Ratio |
|---|---|
| Capital One Financial (COF) | 6.8x |
| CB Richard Ellis (CBG) | 16.2x |
| Genworth (GNW) | 10.4x |
| Assured Guaranty (AGO) | 2.8x |
| Cathay General (CATY) | 8.8x |
| Wilshire Bancorp (WIBC) | 11.8x |
| Jones, Lang Lasalle (JLL) | 2.3x |
| Americredit (ACF) | 5.8x |
| Hertz Global Holdings (HTZ) | 11.4x |

| Portfolio II | Assets-to-Equity Ratio |
|---|---|
| Edwards Lifesciences (EW) | 1.4x |
| Johnson & Johnson (JNJ) | 1.9x |
| Stryker (SYK) | 1.3x |
| Zimmer (ZMH) | 1.3x |
| Sigma Aldrich (SIAL) | 1.7x |
| Precision Castparts (PCP) | 1.4x |
| Analog Devices (ADI) | 1.3x |
| Ebay (EBAY) | 1.4x |
| Barrick Gold (ABX) | 1.5x |

Clearly, Portfolio I is comprised of much more financially leveraged companies than Portfolio II. It is certainly arguable that the businesses of Portfolio II are intrinsically more stable than those of Portfolio I. For example, if I were to compare the business of Sigma-Aldrich, which basically sells unique chemicals used in biopharmaceutical research – not a very variable business – with that of AmeriCredit, which makes sub-prime auto loans, I think most people would agree that Sigma-Aldrich is the more stable business.

The businesses collected in Portfolio I, as a generalization, are far less stable than those of Portfolio II but have far more financial leverage. This is sort of an oddity. The companies that have the most discipline and are the most stable businesses are those that are most able to undertake the risk of leverage. But, in practice, it appears the reverse is true. The businesses that, generally speaking, are less stable, and because of a variety of competitive factors have lower returns on capital invested, have a tendency to leverage more. The reverse should be true, speaking theoretically. Therefore, is it not reasonable to make the assertion that one could create a portfolio of companies similar in character to those listed in this theoretical Portfolio II, and leverage those to a conservative degree with the constraint that one would never exceed the embedded leverage in the S&P 500 itself? Is that not a way of achieving an excess rate of return? I'll allow the reader to ponder these questions.